

Apply for a Science Corps Fellowship - Entry #2417

Name

Samuel Bray

Emailsambray@stanford.edu**Gender**

Male

City, state, and country of residence

East Palo Alto, California, United States

Nationality

American

When did or will you finish your PhD?

December 2022

University Education

Ph.D Bioengineering | Stanford University | 2022

M.S. Bioengineering | Stanford University | 2018

B.S. Bioengineering | University of California, Los Angeles | 2016

What are your reasons for applying to be a Science Corps fellow?

Science is not just to knowing more facts about the world, but to but finding new ways of understanding and seeing it. If our advancement in human intuition is going to keep pace with that of technological capability, we need to move past the barriers that have been built around it. Part of this is developing methods of science communication and teaching that combat the bias that science is a non-artistic, technical process. Part of it is bringing people with perspectives and experiences which have been undervalued in science into the discussion to see things in new ways. If we are to see the world in new ways, we must look at it in directions and from viewpoint in which we have not.

My goal in working within this program would be to develop activities and spaces where science is not just facts in a book but something that can be understood through exploration and play by children who may not see themselves as a 'science person'. Longer term, working with and learning from people with experience in these areas would advance my ability to achieve not just good-willed, but impactful contributions towards creating new opportunities for children to change how they see the world.

In addition to your studies, describe any relevant experiences that have prepared you for a Science Corps fellowship.

One of the most frustrating things I had to learn in grad school was how to communicate a weird idea when words weren't effective. This problem was doubly so when I wanted to share it with family or my partner who all are far from science fields. This led to creativity. Tape, a raspberry camera, and a houseplant turned into timelapses and visualizations of latent properties in machine learning. A set of washers and tubes of paint became a network of pendulums in my shower demonstrating hierarchical patterns in neural dynamics. By bringing scientific ideas to the human scale, I've gained experience in making scientific concepts into human intuition.

On the other hand, one of my favorite parts of grad school was mentoring 5 different undergraduate and high school students. While all of them were quite competent, many came in with no prior research experience. The lack of preconceptions and specific accomplishment requirements let us focus on exploration, which in turn led to co-authored publications. While the amount of protocols in my thesis relying on masking tape, staplers, and jury-rigged imaging chambers is a recurring lab joke, I take pride that something that could be (and was) designed by a teenager can lead to impactful science.

Where do you imagine yourself professionally in the future? Check all that apply.

Academic scientist in academia

Doing science in industry

In science outreach

Expand on your selection above and describe where you imagine yourself professionally in the near (two year) and long-term (ten year) future.

My favorite part of working in science is learning, creating, and demonstrating new ideas. However, my experience of academia leads me to believe that in many cases, long term career positions like professors do not get to engage as directly in these aspects as their students. This is driving me to explore other roles.

Short term, I am just finished defending and looking into roles in non-academic research groups where I can continue to build experience in new areas of research. Long-term my goal is to work in a position where I can continue to be directly involved in research. My experience through grad school has left me with little pretension that massive resources are needed for good science. So whether I end in growing basic science research within industry, a teaching lab in an undergraduate college, or working on exhibits in a science children's museum is something I'm not ready to commit on at this point.

CV/Resume

 [SamuelBray_.pdf](#)

How did you find out about Science Corps?

University email list

Any other comments/questions?

Empty

[Science Corps](#)